

DIGITAL FORENSICS — PRACTICAL EXAMINATION ANSWER KEY

Post Graduate Diploma in Cyber Security and Law

Companion solutions for Practical_Exam_Question_Paper. Each answer gives the terminal method, the GUI (FTK Imager/Autopsy) method where applicable, and the source practical it's drawn from.

Section A — Forensic Imaging & Acquisition

Q1 — Create a forensic image + hash

Terminal:

```
dd if=/dev/zero of=pendrive.img bs=1M count=25
sha256sum pendrive.img
# forensic-grade, one pass:
dc3dd if=pendrive.img hash=sha256 log=pendrive.log
```

FTK Imager (GUI):

```
File → Create Disk Image → Select Source (Physical/Logical/Image File)
→ Add → Image Type: Raw (dd) or E01 → Set destination folder & filename
→ Finish → Start
```

FTK Imager automatically computes and displays the MD5/SHA-1 hash in the imaging summary once complete — this is your baseline for later integrity checks.

Source: 01. Create a Forensics Image for Practice, Physical-Drive-Image-Creation

Q2 — Partition table + mmls/fsstat + Autopsy check

```
dd if=/dev/zero of=pendrive.img bs=1M count=32
parted pendrive.img --script mklabel msdos mkpart primary fat16 1MiB 31MiB
mkfs.fat -F 16 -n "HEXX_USB" --offset 2048 pendrive.img

mmls pendrive.img          # shows the MBR partition table
fsstat -o 2048 pendrive.img # confirms FAT16 filesystem details
```

Autopsy:

```
New Case → Add Data Source → Disk Image or VM File → Browse to pendrive.img
→ Next → Finish
```

In the tree on the left, Autopsy shows a **Volume** node for the partition with its offset and file system type — this should match the offset (2048 sectors) and type (FAT16) reported by mmls/fsstat.

Source: 01.1 Pendrive_Image_Forensics

Q3 — Mount and browse, terminal vs Autopsy

```
mkdir pendrive_mount
sudo mount -o loop,offset=1048576,ro,uid=1000,gid=1000 pendrive.img pendrive_mount/
ls -la pendrive_mount/
sudo umount pendrive_mount/
```

Autopsy: add the same image as a data source (as in Q2) and expand the volume node to browse files in the GUI file tree.

GUI advantage: Autopsy shows deleted files, thumbnails, and metadata inline without extra commands — good for quickly triaging a case.

Terminal advantage: faster for scripting/automation and works over SSH on a headless box with no desktop environment.

Source: 01.1 Pendrive_Image_Forensics, 00-Lab-Setup

Section B — Data Recovery & File Carving

Q4 — Recover a deleted file, terminal + Autopsy, hash match

```
fls -d pendrive.img      # lists deleted entries, e.g. inode 6: secret_notes.txt
icat pendrive.img 6 > recovered_notes.txt
sha256sum recovered_notes.txt
```

Autopsy:

```
Data Source tree → expand volume → "Deleted Files" node
→ right-click the file → Extract File(s) → save to disk
```

Both files should be **byte-for-byte identical**, so their SHA-256 hashes match — proving the deleted data was recovered intact, since deletion on FAT only marks the directory entry free (0xE5); the data blocks remain until overwritten.

Source: 02. Digital_Forensics_Data_Recovery

Q5 — EXIF/GPS metadata, terminal + Autopsy

Terminal:

```
exiftool -GPSLatitude -GPSLongitude -Make -Model -DateTimeOriginal -UserComment vacation_photo.jpg
```

Expected output:

```
GPS Latitude      : 19 deg 4' 33.60" N
```

```
GPS Longitude    : 72 deg 52' 39.72" E
Make            : Apple
Model           : iPhone 14 Pro
Date/Time Original : 2024:03:15 14:23:45
User Comment     : Met with V.S. at the waterfall. Deal confirmed for March 25.
```

Autopsy: select the file in the file list → bottom/right panel → "**Metadata**" tab (or the **EXIF/Media** panel) shows the same GPS/Make/Model/DateTime fields extracted automatically.

Location: GPS 19.0760° N, 72.8777° E corresponds to the **Lonavala area near Mumbai, Maharashtra**.

Source: Internal_Assessment-1_Solutions

Q6 — Carve files with no file system, terminal + Autopsy

```
foremost -t jpg,pdf -i pendrive.img -o recovered/
ls recovered/jpg/ recovered/pdf/
```

Autopsy: in the data source tree, expand "**Unallocated Space**" — Autopsy's ingest modules (File Type Identification + carving) automatically extract embedded JPEG/PDF files found there and list them under "**\$CarvedFiles**".

Why carving works without a file system: header signatures (JPEG FF D8 FF, PDF %PDF-) and footer signatures (JPEG FF D9, PDF %%EOF) are unique byte patterns baked into the file format itself — a carving tool scans raw bytes for these markers and extracts everything between a matched header and the next footer, with no need for any directory/inode structure to say where the file is.

Source: 02. Digital_Forensics_Data_Recovery

Section C — Formatted Drives & File Systems

Q7 — Find hidden files on the evidence image

```
sudo mount -o ro,loop pendrive.img pendrive_mount/
ls pendrive_mount/      # default listing — hidden/dot-files do NOT show

ls -la pendrive_mount/  # -a reveals hidden files (names starting with .)
# or, directly on the raw image without mounting:
fls -a pendrive.img
```

Autopsy: hidden/dot-files appear automatically in the file tree under the data source — no extra step needed, since Autopsy lists every entry regardless of the "hidden" attribute.

A plain `ls` (or Windows Explorer with hidden files off) silently skips any file whose name starts with `.` — a suspect can rename evidence to `.notes.txt` and it simply won't appear. An investigator must always use `-a/-la` (or an Autopsy-style tool that ignores the hidden flag) so nothing on the drive is missed.

Source: 02. Digital_Forensics_Data_Recovery

Q8 — Identify unknown file system by magic bytes

```
xxd -l 512 unknown.img | head
# FAT32 → "FAT32  " label around byte 82
# NTFS  → "NTFS  " label around byte 3
# ext4  → magic 0xEF53 at byte offset 1080 (0x438), stored little-endian as 53 EF

fsstat unknown.img # confirms the type fsstat itself detects
```

Source: Student-Task-02-Disk-Storage-Analysis

Q9 — Timestamp behaviour across media

```
stat original_file.txt      # note Access / Modify / Change times
cp original_file.txt /media/usb/
stat /media/usb/original_file.txt
```

Modified time stays the same (file content is byte-identical), but **Created** and **Accessed** times on the copy reset to the moment of the copy — because the copy is treated as a brand-new file entry on the destination file system, regardless of its content matching the original.

Source: Internal_Assessment-2/Questions

Section D — RAM (Memory) Forensics

Q10 — Capture RAM with FTK Imager

```
FTK Imager → File → Capture Memory
→ Destination path: C:\evidence\memdump.mem
→ ☒ Include pagefile (optional)
→ Capture Memory
```

RAM must be captured **before** shutdown or disk imaging because it is the **most volatile** piece of evidence in the Order of Volatility — running processes, open network connections, encryption keys, and unsaved data all disappear the instant power is lost, while the disk contents remain stable and can be imaged later.

Source: Internal_Assessment-2/Topics, 03. RAM_Volatile_Memory_Forensics

Q11 — Analyse processes in the memory dump

```
vol -f memdump.mem windows.pslist
vol -f memdump.mem windows.pstree
```

Each row reports PID, PPID, ImageFileName, and CreateTime. The applications opened before capture (e.g. notepad.exe, chrome.exe, AcroRd32.exe) appear as their own process entries, typically nested under explorer.exe in the pstree output — confirming they were resident in memory at capture time.

Source: 03.1 RAM Volatility 3, 01-Volatility 3 Installation

Q12 — Analyse network activity in the memory dump

```
vol -f memdump.mem windows.netscan
```

Columns: Proto, LocalAddr:Port, ForeignAddr:Port, State (ESTABLISHED/CLOSED/LISTENING), PID, Owner. Match the PID/Owner column back to the process list from Q11 to identify which application owned each connection. `netscan` can recover connections that were already closed by capture time, since it scans the memory pool directly rather than relying on a live connection table.

Source: 03. RAM_Volatile_Memory_Forensics

Section E — Log Analysis & Full Investigation

Q13 — Web server directory scan detection

```
grep -E "DirBuster|Nikto|dirbuster|gobuster" /var/log/apache2/access.log
```

Attacker IP: 185.220.101.5

Signature: User-Agents Nikto/2.1.6 and DirBuster-1.0-RC1

Time window: 01/Apr/2026 02:00:10 → 02:00:40 — dozens of requests to `/DVWA/`, `/phpmyadmin/`, `/wp-admin/`, `/admin/`, `/backup/`, `/uploads/`, `/includes/`, `/config/`, `/db/` in seconds, mostly returning 403/404.

Source: Log_Analysis_Digital_Forensics, forensics-lab-Logs/

Q14 — SSH brute-force detection

```
grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -rn
```

Attacker IP: 185.220.101.5

Evidence: repeated Failed password for root, then admin, at roughly one attempt per second starting Apr 1 02:00:01 — many different usernames tried in rapid succession from a single source IP is the classic brute-force signature (compare to the earlier 172.16.0.5 entries, which show only 2 failed attempts before the connection closed — reconnaissance, not a sustained attack).

Source: Log_Analysis_Digital_Forensics, forensics-lab-Logs/

Q15 — Complete investigation (capstone)

1. Hash and mount read-only

```
sha256sum Devid_evidence.img  
sudo mount -o ro,loop,offset=1048576 Devid_evidence.img /mnt/evidence
```

Also open the image as a **new case in Autopsy** (Add Data Source → Disk Image or VM File) alongside the terminal work.

2 & 3. File system, partition layout, deleted files

```
mmls Devid_evidence.img  
fsstat -o 2048 Devid_evidence.img  
fls -d -o 2048 Devid_evidence.img  
icat -o 2048 Devid_evidence.img <inode> > recovered/btc_transfer_record.txt
```

Or recover the same files via Autopsy's "**Deleted Files**" view.

4. Metadata from recovered photo

```
exiftool /mnt/evidence/vacation_photo.jpg
```

Or via Autopsy's **Metadata** tab on the file.

5. Timeline

```
fls -o 2048 -m / Devid_evidence.img > bodyfile.txt  
mactime -b bodyfile.txt -d > timeline.csv
```

Or use Autopsy's built-in **Timeline** feature (bottom toolbar) for the same MAC-time view in a GUI.

6. Summary (example):

The 200 MB FAT32 image contained four visible files and four deleted files, all recovered successfully via both Sleuth Kit and Autopsy. The recovered photo vacation_photo.jpg carried GPS coordinates placing the device near Lonavala on 2024-03-15, corroborated by a recovered text file referencing a "deal confirmed for March 25" — consistent with the timeline showing all suspicious files created and deleted within the same session.

Source: Student-Task-09-10-OS-Artifacts-Investigation, Internal_Assessment-1_Solutions